

DevOps/AWS Engineer

Akanksha CH

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Professional Summary:

- Having 5+ years of IT Professional Experience as a DevOps Engineer comprising of Linux and System Administration with a major focus on AWS, Open Stack, Continuous Integration, Continuous Deployment, Configuration Management, Build/Release Management, and Virtualization technologies which also includes Troubleshooting and Performance issues.
- In-depth experience with SaaS, PaaS, and IaaS concepts of cloud computing architecture and Implementation using AWS, Google Cloud Platform, OpenStack, Pivotal Cloud Foundry (PCF), Slack, and Salesforce.
- In-depth understanding of the principles and best practices of software configuration management (SCM) in Agile, Scrum, and Waterfall methodologies.
- Experience in Designing, configuring, and managing public and private cloud infrastructures utilizing Amazon Web Services (AWS) like Elastic Container Service (Docker Containers), S3, Elastic Beanstalk, Cloud Front, Elastic File System, RDS, DynamoDB, Redshift, DMS, Kinesis, VPC, Direct Connect, Route53, Cloud Watch, Cloud Trail, Cloud Formation, IAM, EMR and Elastic Search.
- Experience with AWS Command Line (CLI), AWS SDK, current AWS product offerings and services, best practices and troubleshooting techniques, and experience providing systems administration and issue resolution for AWS.
- Having good knowledge of Google Cloud Platform (GCP) and expertise in Object Oriented Programming using core Java, Groovy, and J2EE.
- Expert in designing Continuous integration and developing Jenkins and Anthill, performed end-to-end automation for build and deployments.
- Expertise in Branching, Merging Strategies, and migrating projects from Subversion repositories to GitHub Enterprise repositories through the Team Foundation server (TFS).
- Hands-on in using Bamboo modules such as BuildCompleteAction, BuildPlanDefinition, and Administration configuration.
- Experience in Writing Puppet manifests for deploying, configuring, installing shields, and managing collected for metric collection and monitoring and constructing the Puppet modules for continuous deployment.
- Expertise in using Docker to containerize custom web applications and deploy them on Digital Ocean with Ubuntu instances through SWARM Cluster and to automate the application deployment in the cloud using Vagrant.

- Expertise in implementing a production-ready, highly available, fault-tolerant Kubernetes infrastructure and Worked on Scheduling, deploying, and managing container replicas on a node cluster using Kubernetes.
- Expertise in using build tools like MAVEN, ANT, and Gradle for building deployable artifacts such as war and ear from source code.
- Experience in working with Application and Web servers like IIS, Tomcat, Apache, Confidential, JBoss, WebLogic, Nginx, Open VPN, Sensu, log stash, and in Design patterns such as Model View Controller (MVC), Strategy Pattern, Command Pattern, Factory Pattern, Inversion of Control and Singleton Pattern.
- Experienced in all phases of the Software Development Life Cycle (SDLC) with a specific focus on the build and release and quality of software and involved with teams that worked on Scrum, Agile Software Development, and Waterfall Methodologies.
- Excellent experience in designing, documenting, and automating the build and release process and Good analytical, presentation & and communication skills and ability to adapt to new technologies and project environments.

Technical Skills:

Cloud Computing: Amazon Web Services (EC2, IAM, Elastic Beanstalk, and Elastic Load balancer (ELB), RDS (MySQL), S3, Glacier, Route 53, SES, VPC, etc.), Open Stack Platform Cloud Foundry.

Configuration Management: ANT, Maven, GIT, SVN, Clear Case, Jenkins, Puppet, Chef, Ansible, Sonar, Nexus.

Tools/ Webservers: Web Sphere Application Server 3.5, 4.0, Web Logic, JBoss, Apache, Tomcat, Nagios, Kafka, Log stash, Spark

Scripting/ Languages: Shell scripting, Python, Ruby, and PowerShell Scripting.

Database: Sybase, Oracle 7.x/8.0/9i/10g/ 11g, MySQL, DB2, Angular JS, Cassandra, Mongo DB

Networking/ Protocols: DNS, TCP/IP, FTP, HTTPS, SSH, SFTP, SCP, SSL, ARP, DHCP and POP3

Operating Systems: Sun Solaris 7, 8, 9 &10 Linux (Red Hat 5. x, 6. x, 7. x' SUSE Linux 10), AIX, VMware ESX, Windows NT/ 2000/2003/2012, Centos, Ubuntu.

Education:

Masters from Troy University – Troy, Alabama, 36081

Professional Experience:

Client: Dew Software, Fremont, CA

May 2023 - Present

AWS DevOps Engineer

Responsibilities:

- Developed environments of different applications on AWS by provisioning EC2 instances and configuring Terraform, Bash, and Docker.
- Created AWS Instances, Instance Groups, AWS IAM, and Security Groups in Public and Private Subnets in VPC.

- Configured multiple Elastic Load Balancers (ELB) with EC2 Auto Scaling Groups for high availability; fault-tolerant Docker containers in ECS.
- Developed automated provisioning applications according to organizational standards for deployment to the AWS environment.
- Built Jenkins jobs to create AWS infrastructure from GitHub repos containing terraform code. Hands-on experience on Terraform as a tool for building, changing, and versioning infrastructure safely and efficiently.
- Maintained the cloud infrastructure using a combination of Jenkins, and Terraform for automating the CICD pipeline in AWS.
Worked with the application team on testing and validating the new deployments.
Configured Keyblade tool for application configurations.
- Configured RDS, and Elastic Cache and encrypted the security credentials in corresponding VPC configuring Vault.
- Extensively worked on Vault Secret Management and Consul tools to provide security credentials, tokens, and API keys.
- Configured RDS, and Elastic Cache and encrypted the security credentials in corresponding VPC configuring Vault.
- Created AWS lambda batch Jobs and scheduled jobs based on the company requirements. Designed and worked with the team to implement ELK (search, log stash, and Kibana) Stack on AWS.
- Migrated legacy applications and underlined platforms from data centres to AWS cloud. Experience with several Docker components like Docker Engine, Hub, Machine, Compose, and Registry.
- Experienced in creating RDS instances to serve data through servers for responding to requests. Configured Multi-AZ RDS instances for High Availability and Read Replica instances for serving read-intensive applications.
- Experience in configuring Splunk and Integrated Splunk with Active Directory and LDAP authentication.
- Performance monitoring of CPU, Memory, Disks, and network of multiple services configuring AS Cloud Watch metric logs.
- Experience with analysis of logs and performing root cause analysis for various issues coming up once the application went to the production region.
- Experience in deploying the application in multiple Testing and QA environments.
Worked with different team members for automation and Release components.

Environment: AWS, EC2, AMI, S3, EBS, Elastic Load balancer (ELB), Auto Scaling groups, Glacier, VPC, IAM, Cloud Watch, Akamai, DynamoDB, MySQL, shell scripts, Elasticsearch, Logstash, Kibana, Git, GitHub Maven, Chef, Artifactory, Selenium, Docker, Mesos, Jenkins, uDeploy, PowerShell.

Cloud Engineer (Intern)

Responsibilities:

- Being an active role in the application infrastructure management team, I was completely involved in setting up the new environment for Continuous Integration (CI) and Continuous Deployment (CD) in Agile software methodology.
- Evaluated architecture proposals for data migration and used Infrastructure as code with almost Zero downtime and designed a serverless data ingestion pipeline leveraging multiple services, which was the most cost-efficient solution that reduced workload by 30% and was affordable for a DevOps team to execute and operate.
- I have migrated data from the on-premises data centre to the AWS cloud using the AWS snowball import export device.
- Responsible for Multiple AWS instances, setting the security groups, Elastic Load Balancer and AMIs, and Auto scaling to design cost-effective, fault-tolerant, and highly available systems.
- Created Lambda Scripts to Automate AWS services, including web servers, ELB, Cloud Front Distribution, database, EC2, and database security groups, S3 bucket, and application configuration, this Script creates stacks, single servers, or joins web servers to stacks and designed custom scripts for running cron jobs.
- Wrote Python Program on Lambda to Ingest Raw JSON data received through API Calls into the Amazon Kinesis Streams Used Cache DB-based Redis Server to perform realtime data ingestion traceability.
- Performed Data Validations and transformations which included filtering, mapping data fields, and converting data formats Using Elastic Map Reduce (EMR) and Kinesis.
- AWS EC2 and Cloud watch services, CI/CD pipeline management through Jenkins automation of manual tasks using Shell scripting.
- Provisioned the highly available EC2 Instances using Terraform and cloud formation and wrote new plugins to support new functionality in Terraform.
- Deployed and configured Elastic Search, Log stash, and Kibana (ELK) for log analytics, full-text search, and application monitoring in integration with AWS Lambda and Cloud Watch.
- Creating, versioning, and testing scripts (Bash, PowerShell), Chef, Nagios, Maven, Git, Jenkins, Python, and Ruby to achieve a high level of automation.
- Built end-to-end CI/CD Pipelines in Jenkins to retrieve code, compile applications, perform tests, and push build artifacts to Nexus and uDeploy to orchestra changes across servers and components.
- Well-versed in Groovy and Js script used mostly for automation scripting purposes and release modules using Configuration Management.
- Worked with Version control tools Subversion/Git Branching, Merge Strategies, and migrated projects from Subversion repositories to GitHub Enterprise repositories.
- Worked on User administration setup, maintaining accounts, and monitoring system performance using Nagios and Splunk.

Environment: AWS, Terraform, GCP (Google cloud platform) Cassandra, Kafka, Zookeeper, IIS, Bash, Lambda, Python, Chef, Docker, PCF, Kubernetes, Jenkins, Linux, Maven, Ant, JIRA, Agile, Nagios, Vagrant, SVN, ELK, ARM, Perl, JSON, Ruby, Splunk, PowerShell, Bitbucket, Git.

Client: Wells Fargo, CA

June 2019 – Sep 2021

DevOps Engineer

Responsibilities:

- Developed build and deployment processes for Pre-production environments.
- Used Subversion as source code repositories. Developed Shell/Python Scripts for automation purposes.
- Managing a team of DevOps engineer for infrastructure support on AWS cloud Creating cloud formation scripts for hosting software on AWS cloud and automating the installation of software through PowerShell scripts.
- Managed multiple AWS accounts with multiple VPC's for both production and non-prod where primary objectives included automation, build-out, integration, and cost control.
- Designed IaaS and PaaS solutions for new clients migrating from onsite infrastructure to the cloud.
Built Jenkins job setup for applications using PowerShell scripting and used ANT and MAVEN as build tools on Java projects for the development of build artifacts on the source code.
- Migrated ANT projects into Maven projects and wrote Maven POM files from scratch for multi-module projects.
- Resolved merging issues during rebasing and re-integrating branches by conducting meetings with Development Team Leads.
- Ability to build deployment, scripts, and automated solutions using various scripting languages such as Shell, PowerShell, Python, and Ruby.
- Created and wrote shell scripts Bash, Ruby, Python, and PowerShell for automating tasks. Automated Weekly releases with ANT and Maven scripting for Compiling Java Code, Debugging, and Placing Builds into the Maven Repository.
- HTTP/1.1 to HTTP/2.0 for better connection management. Built, implemented, and maintained hardware and software infrastructure to host TFS as a SaaS solution including.
- Responsible for designing and deploying the best SCM processes and procedures. Used Ansible to automate Configuration management and to manage Web Applications, Configured Files, Data Base, Commands, user mount Points, and Packages.

Environment: AWS, AIX and Windows, PCF, BMC Remedy, Shell, Python, Java, Web Logic Server 11g, Active Directory, JDK1.7, XML, Anthill Pro, SVN, Jenkins, Ansible, Maven, GIT, Jira, Docker, SQL.